

Advantages of Quantum Recurrent Networks: An Examination of Stable States

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Abstract – We continue our investigation of the ways in which both quantum computing and neural computing can be augmented by combining the two. Here, we quantify the increase in storage capacity of the quantum Hopfield network, as a function of its increasing quantum mechanical nature.

I. INTRODUCTION

Today's digital computers are based on the concept that information storage, manipulation and retrieval are in the form of discrete value known as digital bit. Conceptually the bit has either one or zero in magnitude but physically it is represented by 5 and 0.5 volts signal in most of the cases. Quantum computation and quantum computers, currently being researched, use the quantum bit, or qubit, that has a logic value, such as up or down, 0 or 1, or +1 and -1, but can also take on a continuum of values between up and down, i.e. $[0 \rightarrow 1]$, or $[+1 \rightarrow -1]$. In other words, unlike digital circuits, the logical value may be more up than down, or more down than up. These in-between states are called superposition states, as they are a superposition of an up and a down state. This new concept of continuous bit allows a wider range of information storage and processing.

Traditional Hopfield networks have been shown to be able to store information as stable states of the recurrently connected network. These states exist at the corners of the n -dimensional hypercube, where n is the number of neurons in the network. Due to the superposition states available in a quantum recurrent Hopfield network, many many more stable storage states are available to this quantum network. An investigation into this increase in storage capacity is the major contribution of this paper.

The Hopfield neural network is a model of associative content-addressable memory; it is capable of storing information as well as carrying out certain computational tasks such as error correction, etc. In Hopfield neural networks the processing units are fully interconnected, i.e., each unit is connected to every other unit. Hopfield [1] demonstrated that associative memory can be implemented with his network and optimization tasks such as the traveling salesman problem can be solved. A Hopfield network is initialized to a beginning state. Then each neuron's output is updated according to its weighted inputs. The updating of the neurons continues until the network converges to a stable state, or fixed point.

A **binary** Hopfield network has neurons which can be in one of two states: $S_i = +1$ (on) and $S_i = -1$ (off), where i is the number of the neuron. Each neuron is synaptically connected

to other neurons with multiplicative weights that are real numbers. Each neuron is randomly selected and sums up its input and decides whether to turn itself on or off. Let W_{ij} be the synaptic strength from neuron j to neuron i and let I_i be the external input to unit i . Each neuron takes a weighted sum of its inputs according to the following equation:

$$S_i = \text{sgn} \left(\sum_{j \neq i}^n W_{ij} S_j + I_i \right) \quad (1)$$

After repeated updating, the network may achieve a stable point, meaning, in the state space, all neurons remain in their current state after examining the inputs. The output is a stable vector-of-states which constitutes the recall of a stored pattern in the memory.

In the **continuous** Hopfield net, updates in time are continuous, rather than discrete. The governing equation of the i^{th} neuron is

$$\begin{aligned} \frac{du_i}{dt} &= \sum_j W_{ij} S_j - \frac{u_i}{R_i} + I_i \\ S_j &= \tanh(\gamma u_j) \end{aligned} \quad (2)$$

where R_i is the decay rate ($R_i > 0$) and S_j is the output of unit j after the activation function $\tanh$ is applied. γ is the gain of the activation function. The Hopfield network considered in this paper is in the high gain limit where γ and R_i are large, and the $\tanh$ approximates the sgn function of the discrete network.

Each state in the high gain Hopfield neural network has a Hopfield "energy" or "Lyapunov" function, given by

$$L = -\frac{1}{2} \sum_i \sum_j W_{ij} S_j S_i - \sum_i S_i I_i \quad (3)$$

L is a Lyapunov function for the dynamics of equations (2) since it is bounded and it can be shown [2] that $dL/dt < 0$ except at the fixed points. Thus the Hopfield network is globally asymptotically stable and the stable states are the minima of the "energy" function L . The processing of the Hopfield neural network, from an initial to a final steady state, is a process by which the "energy" of the system becomes smaller and eventually achieves a local minimum at the stable state. The weights, the external inputs, and the initial state determine which stable state, or pattern, is recalled by the network.

II. QUANTUM HOPFIELD NETWORK

The simplest quantum system is a two level system (TLS). A TLS has two distinct physical states, switching from one to another under certain conditions. Because of quantum mechanical tunneling there is always some finite probability that the system may tunnel through a potential barrier – that is, change states without having, classically, enough energy to do so. In alpha decay, alpha particles tunnel through the strong nuclear potential barrier and get out of the nucleus. Similarly an electron can barrier tunnel through an oxide layer in a MOSFET component. We can envision this as applying an electric field across the layer, so an electron that is trapped in one of the separated potential wells tunnels to the other. Classically we assign one well to represent the 0 bit and the other the 1 bit. But in a quantum mechanical system, because of tunneling, the electron can be simultaneously in both states – in a superposition of states. Because the superposition is a continuum so are the possible states of the quantum bit or qubit.

The quantum system can be put into the Hopfield form to obtain the quantum Hopfield net (QHN) as follows. We write the partition function in the Feynman path integral formulation [3] for the array of N qubits as:

$$Q = \int \prod_{j=1}^N D\sigma_{j\cdot}(t) \exp \left\{ - \int_0^\beta dt \sum_{j=1}^N K \sigma_{j\cdot}(t) + \int_0^\beta dt \int_0^\beta dt' \sum_{j \neq j'=1}^N \sigma_{j\cdot}(t) \alpha_{jj'}(t, t') \sigma_{j'\cdot}(t') \right\} \quad (4)$$

where the script D indicates the summation over all possible paths (the “path integral”). Here β is the inverse temperature in units of Boltzmann’s constant, α represents the interaction between qubits, and the state of the qubit is represented by the Pauli operator or matrix σ_z for each qubit j . (This is equivalent to the more usual representation of the partition function as $Q = \text{tr} \exp\{-\beta H\}$, where H is the Hamiltonian and ‘tr’ refers to the trace. The Hamiltonian gives the energy of the system. The path integral formulation is sometimes referred to [4] as propagation in imaginary time, because the Boltzmann factor $\exp\{-\beta H\}$ is the same as the expression for time evolution $\exp\{-itH/\hbar\}$, for $t = -i\beta\hbar$, with $\hbar$ Planck’s constant divided by 2π .) The first term in the exponential in equation (4) is the tunneling term, containing the amplitude for tunneling, K , multiplied by the Pauli matrix σ_x . The second term contains the interactions between each pair of qubits j, j' , with an interaction strength $\alpha_{jj'}$ which may, in general, depend on both the identities of the interacting qubits and their separation in imaginary time. Because the partition function is a trace the state of each qubit at “time” 0 is constrained to be equal to its state at “time” β , so the state of each qubit can

be thought of as a loop. The path integral is equivalent to an infinite number of sums as follows:

$$Q = \lim_{n \rightarrow \infty} \sum_{S_j = \pm 1} \exp \left\{ \zeta \sum_{j=1}^N \sum_{t=0}^{n-1} S_j(t) S_j(t+1) + \sum_{j \neq j'=1}^N \sum_{t=0}^{n-1} S_j(t) \alpha_{jj'}(t, t') S_{j'}(t') \right\} \quad (5)$$

where the functions $\{S_j\}$ are the instantaneous values assumed by each of the qubits j , at each imaginary time t along the inverse temperature loop, and the first term accounts for the tunneling of the qubit, where

$$\zeta = -\frac{1}{2} \ln [\tanh(\beta K / n)] \quad (6)$$

For full details see [3], [4], and [5]. The discretized representation is exact in the limit that n is infinite. The continuum of possible states of each qubit is represented by a continuum of bits. For finite n we can compute the state of the net exactly by summing over all possible states of each of the discretized points of each of the qubits.

Qualitatively, the imaginary time (inverse temperature) loops represent the quantum dispersion, or spreading out of the wavefunction. At low temperature (the quantum limit) β is large, so each qubit has a great deal of freedom (a long imaginary time) before it is constrained to return to its original point. In this limit a very large number n will be necessary to represent the dispersion. At higher temperature β becomes smaller, until in the classical limit the dispersion loop shrinks to a single point (i.e. $n=1$). Physically the qubit is no longer able to tunnel. The (finite) value of n we use will be large enough (the approximation will be good) if $\beta H/n$ is small.

Each set of instantaneous values of each of the qubits, $\{S_j(t)\}$, can be thought of as a microstate. The contributions from all of the possible microstates are summed over to obtain the partition function. The contribution from a microstate to the partition function is large if its associated energy is low. Thus, states that are more probable can be found by maximizing the argument of the exponential in equation (5) – or, since each exponential corresponds to a Boltzmann factor $\exp(-\beta E)$, minimizing its associated energy. We rearrange that argument to correspond with equation (3); that is, we define the Lyapunov function for the QHN as the integral of the Hamiltonian over the imaginary time loop, divided by β – i.e., the quantity in curly brackets in equation (5). This is the QHN. Classically, the high gain Hopfield net replaces a continuum of possible values of each neuron with a finite number; in the QHN it becomes again a continuum.

III. STABLE STATES OF THE NETWORK

Information in the Hopfield net is stored as stable states. Classically the discrete net converges to the corners of an N -

dimensional box [2], where the neuron outputs necessarily equal plus or minus 1. In a network of N classical neurons there are 2^N possible states (corners of the box.) Once we discretize the quantum Hopfield net, to discretization number n , there are 2^{Nn} possible states: The greater the discretization, the greater the number of possible states. When the temperature is lower we need a larger discretization number; thus, the lower the temperature the greater the storage capacity of the QHN. For infinite n we have the fully quantum mechanical system, and have a fully faithful representation of the quantum dispersion - all possible superposition states.

Are all the corners of box are local minima or stable states? If so, we have an infinite storage capacity for the QHN: The $N \times n$ dimensional cube has become a sphere, and any point on the surface of a sphere is a "corner." To answer this we consider a QHN at finite n , bearing in mind that only $n=1$ (classical) and $n=\infty$ (fully quantum mechanical) have any recoverable physical meaning; still, " $n=\infty$ " should be interpreted as equivalent to $\beta H/n \ll 1$. Figure 1 shows a picture of the net for $N=2$ and $n=2$. Each (full-line) loop represents a qubit or TLS, propagating in imaginary time (inverse temperature), from $t=0$ to $t=\beta$. Each dot on a loop represents the instantaneous state of the qubit at a particular value of t , and these dots interact with their nearest neighbors along the loop (the first term in equation (5).) This is the tunneling term. The arrows show the direction of integration, from $t=0$ to $t=\beta$. Each dot also interacts with other dots on other loops according to the second term in equation (5). In the simplest possible model, which we consider here, the two qubits interact only at equal imaginary times t , and the Lyapunov function from equation (5) becomes:

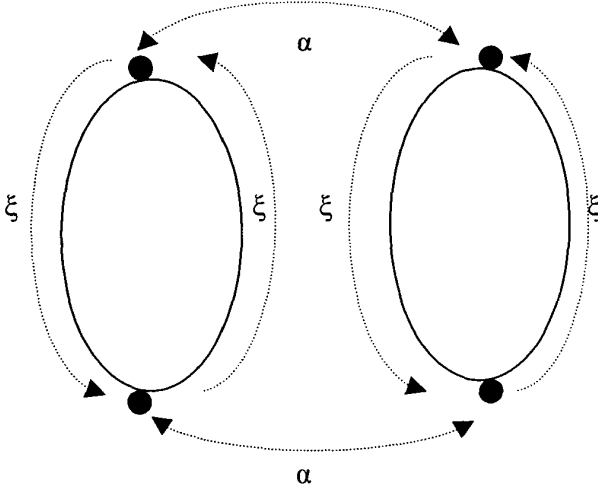


Figure 1: Representation of two interacting qubits, with discretization $n=2$. Each loop represents a qubit, propagating in imaginary time (inverse temperature) from 0 to β . The interactions between discretization points on the same loop have a strength of ξ ; the (bidirectional) interactions between points on different loops, of α .

$$L = -\frac{1}{\beta} \left[\zeta \left(\sum_{i=1}^n S_i S_{i+1} + \sum_{i=1}^n S'_i S'_{i+1} \right) + \alpha \left(\sum_{i=1}^n S_i S'_i \right) \right] \quad (7)$$

where the (primed, unprimed) S functions refer to the two interacting loops.

As mentioned earlier all the extrema are in the corners. These can be minima, maxima or saddle points. The Hopfield algorithm guarantees a minimum, or at least a relative minimum (subject to the constraint that each S has absolute value 1.) We can see that this is true by putting equations (2) and (3) together, to obtain

$$\begin{aligned} \frac{dL}{dt} &= \sum_i^n \frac{\partial L}{\partial S_i} \frac{dS_i}{dt} = \sum_i^n \left[-\sum_j^n W_{ij} S_j - I_i \right] \frac{dS_i}{dt} \\ &= -\sum_i^n \frac{du_i}{dt} \frac{dS_i}{dt} = -\sum_i^n \frac{du_i}{dt} \frac{dS_i}{du_i} \frac{du_i}{dt} \\ &= -\sum_i^n \frac{dS_i}{du_i} \left(\frac{du_i}{dt} \right)^2 \leq 0 \end{aligned} \quad (8)$$

since the tanh is a strictly increasing function. We can analyze the Lagrangian in equation (7) to identify the stable points of the quantum Hopfield net. We do this by minimizing L subject to the constraint that each S must have absolute magnitude less than or equal to one using the method of Lagrange multipliers[6]. (The tanh substitution does this automatically for the Hopfield net.) We can define a new functional

$$F \equiv L + \sum_i^n l_i h_i \quad (9)$$

where $h_i = (S_i^2 - 1)$ and l_i are the Lagrange multipliers, each multiplying a quantity h_i that is constrained to be less than or equal to zero. The Kuhn-Tucker condition then tells us that there is a vector $\mathbf{l}$ of multipliers, such that $\nabla F = 0$ and $\mathbf{l} \bullet \mathbf{h} = 0$, where $\mathbf{h}$ is the vector of conditions and ∇ is as usual the gradient (with respect to each of the S variables).

By applying above mentioned condition on our system we have an algorithm through which we can find the number of stable states as a function of n . Figure 2 shows results for the uncoupled system (i.e., $\alpha=0$). We see that the number of stable states does increase exponentially but not as fast as the total number of states, which goes as 2^n . The slope of the line, as n becomes large (the quantum limit), is 0.48. Because this is an isolated system the value of the slope is not a function of the inverse temperature β or the tunneling amplitude K . In contrast, the corresponding classical net has $\ln(\#states) = \ln(2) = 0.69$, for all values of n .

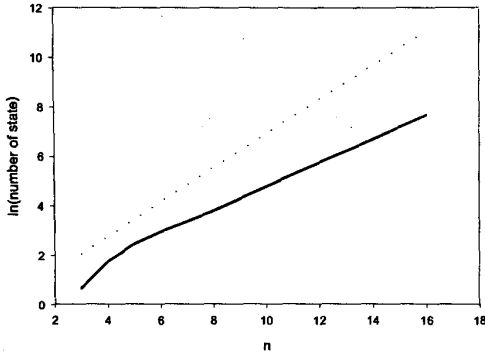


Figure 2: Natural logarithm of the number of states, as a function of quantum discretization number n , for the uncoupled system. The full line is the number of stable states of the QHN; the dashed line is the total number of possible states, which goes as 2^n .

Figure 3 shows results for non-zero α . (Note the change in scale of the abscissa.) Here $\beta=2.0$, $K=0.1$, and $\alpha=1.5$. This is a value for α at which the contributions to the Lyapunov function from the two terms are approximately equal. The slope of the line, again in the quantum limit, is now equal to 0.51. The increase in the slope means that the storage capacity of the net, at any given value of n , is greater with interactions between the qubits than without. For smaller but still nonzero α the slope does not appear to change.

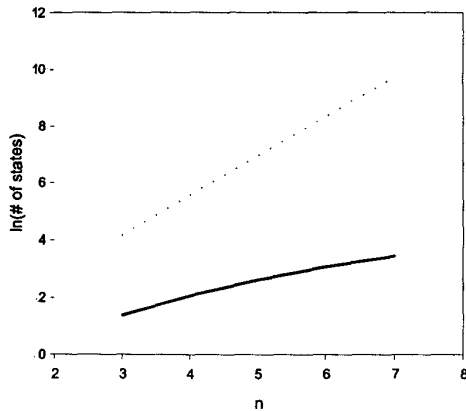


Figure 3: Natural logarithm of the number of states of the QHN, as a function of quantum discretization n , for $\beta=2.0$, $K=0.1$, and $\alpha=1.5$. Again the dashed line shows the number of possible states and the full line shows the stable ones.

Finally we present results for α large enough that its effects dominate the physics of the system. Figure 4 shows results for $\alpha=7.0$. Again the slope has increased, to 0.69.

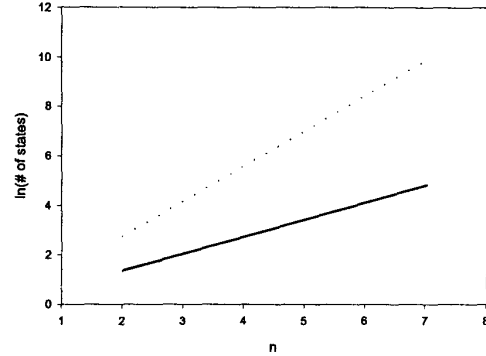


Figure 4: Natural logarithm of the number of states, for $\alpha=7.0$. Again the dashed line is the total number of possible states, for comparison. The other parameters were the same as in Figures 2 and 3.

IV. HOPFIELD SIMULATIONS

We have also performed Hopfield simulations of the QHN. Equation (2) for the QHN is:

$$\frac{du_i}{dt} = \frac{2\xi}{\beta} (S_{i-1} + S_{i+1}) + \alpha S_i - \frac{u_i}{R_i} + I_i \quad (10)$$

$$S_j = \tanh(\gamma u_j)$$

We start the $\{S_j\}$ out in some initial state, with $-1 \leq S_j \leq +1$, having chosen values for the physical parameters ξ , α , I_i , and R_i , and for the gain γ . We then iterate equation (10) for each of the $n \times N$ $\{S_j\}$ to a steady state. Each steady state is a local minimum of the Lyapunov function, which is the energy of the associated physical system. All the states predicted to be minima by the method of section III are steady states of equation (10). Some results are shown in Table 1. This again is in the high-gain limit. Table 1 shows all possible states for $N=2$, $n=3$, of the QHN, for the physical system of Figure 3, i.e., for small nonzero couplings of $\alpha=1.5$ and $\alpha=7$, as calculated by the methods of Sections III and IV (which agree.) The Hopfield parameters used were: $\gamma=10$, $R=106$, $I=0.0$, and $\Delta t=10^{-3}$. The highlighted states are the stable ones, labeled "S" in the seventh and ninth columns, for $\alpha=1.5$ and $\alpha=7$ respectively. Columns eight and ten also show the values for $-\beta E$, i.e., the negative of the energy of the state in units of kBT, for $\alpha=1.5$ and $\alpha=7$ respectively.

						$\alpha=1.5$		$\alpha=7$	
S1	S2	S3	S'1	S'2	S'3	S/U	-BE	S/U	-BE
1	1	1	1	1	1	S	11.59	S	29.13
1	1	1	1	1	-1	U	3.86	U	9.71
1	1	1	1	-1	1	U	3.86	U	9.71
1	1	1	1	-1	-1	U	1.56	U	-4.29
1	1	1	-1	1	1	U	3.86	U	9.71
1	1	1	-1	1	-1	U	1.56	U	-4.29
1	1	1	-1	-1	1	U	1.56	U	-4.29
1	1	1	-1	-1	-1	S	4.67	U	-12.87
1	1	-1	1	1	1	U	3.86	U	9.71
1	1	-1	1	1	-1	U	0.75	S	18.29
1	1	-1	1	-1	1	U	-3.86	U	-9.71
1	1	-1	1	-1	-1	U	-1.56	U	4.29
1	1	-1	-1	1	1	U	-3.86	U	-9.71
1	1	-1	-1	1	-1	U	-1.56	U	4.29
1	1	-1	-1	-1	1	U	-6.17	U	-23.71
1	1	-1	-1	-1	-1	U	1.56	U	-4.29
1	-1	1	1	1	1	U	3.86	U	9.71
1	-1	1	1	1	-1	U	-3.86	U	-9.71
1	-1	1	1	-1	1	U	0.75	S	18.29
1	-1	1	1	-1	-1	U	-1.56	U	4.29
1	-1	1	-1	1	1	U	-3.86	U	-9.71
1	-1	1	-1	1	-1	U	-6.17	U	-23.71
1	-1	1	-1	-1	1	U	-1.56	U	4.29
1	-1	1	-1	-1	-1	U	1.56	U	-4.29
1	-1	-1	1	1	1	U	1.56	U	-4.29
1	-1	-1	1	1	-1	U	-1.56	U	4.29
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1	-1	-1	-1	1	-1	U	-3.86	U	-9.71
1	-1	-1	-1	-1	1	U	-3.86	U	-9.71
1	-1	-1	-1	-1	-1	U	3.86	U	9.71

-1	1	1	1	1	1	U	3.86	U	9.71
-1	1	1	1	1	-1	U	-3.86	U	-9.71
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-1	1	-1	1	1	-1	U	-1.56	U	4.29
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-1	-1	-1	-1	1	1	U	1.56	U	-4.29
-1	-1	-1	-1	1	-1	U	3.86	U	9.71
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-1	-1	-1	-1	-1	-1	S	11.59	S	29.13

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V. CONCLUSIONS AND FUTURE WORK

We have shown that the QHN has a storage capacity surpassing that of the classical Hopfield net, though not as great as the discretization implies. Clearly more investigation is needed. How the slope of the line depends in detail on the relative size of α and ζ is a question we mean to answer. Our major avenues of research at the moment are concerned with the use of entanglement in the QHN and with the training of the QHN. Preliminary work [5] shows that it can indeed be trained – that the relative depths of minima can be altered by adjusting externally accessible parameters. Experimental accessibility is a separate question. The individual "dots" (quantum discretization points) are not addressable. However the

continuum nature of each qubit is, so this is not a problem. Somewhat more worrisome is the question of reading and writing the data.

VI. ACKNOWLEDGEMENTS

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